



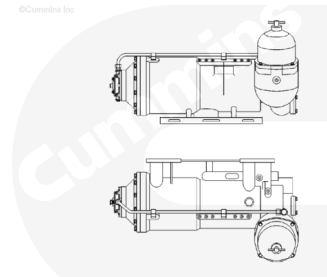
Service Bulletin Number: 3666253

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Eliminator™ Filter

Eliminator™ Filter

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General Information

Safety Instructions

Pay attention to the safety instructions in this bulletin.

Shown are two different types of safety instructions used in this bulletin.

⚠ WARNING ⚠

This type of instruction is used where serious injury or extensive property damage can result if the warning instructions are not followed.

⚠ CAUTION ⚠

This type of instruction is used where minor personal injury can result or a part, an assembly, or the engine can be damaged if the caution instructions are not followed.

On the following pages you will find:

- Warning Signs and Labels
- Safety Precautions and Instructions. These contain important information concerning safety.

Warning Signs and labels

⚠ WARNING ⚠

FAILURE TO FOLLOW THE SAFETY INSTRUCTIONS CAN CAUSE ACCIDENTS RESULTING IN SERIOUS INJURY TO PERSONS AND DAMAGE TO THE EQUIPMENT.

Safety Precautions and Instructions

Incorrect installation, operation, and maintenance, or use of non OEM parts, can cause a breakdown of the equipment.

Personnel installing, operating or performing service **must** have proper knowledge and skill for their task. **It is essential to read, understand and follow the instructions in this bulletin.**

- **Always** follow safety regulations, local and general, concerning lifting devices, electrical equipment, pressure vessels, inflammable, toxic, or corrosive process media, etc.
- **Always** follow Cummins Inc. specifications concerning checking and maintenance intervals.
- **Always** follow stated values for process parameters.
- **Always** use ear protection when working in noisy environments.
- **Never** use the filter for filtering other liquids than originally specified. If your requirements change, **always** consult a Cummins® Authorized Service representative before any changes are made.
- A filter has minimum and maximum safe operating limits concerning temperature, pressure, flow, etc. (see Technical data and drawings). Operating above or below these limits may be potentially dangerous.
- Be careful when working near pipes, valves, heaters, or motors. If you come in contact with hot liquids, hot surfaces or steam, you can experience severe burns.

Note : If you are uncertain about any point, contact a Cummins® Authorized Service Representative.

Working Practices

Good working practices help to avoid unnecessary injury to personnel or damage to equipment and can prolong the working life of the filter. The following points will contribute towards this:

Warranty

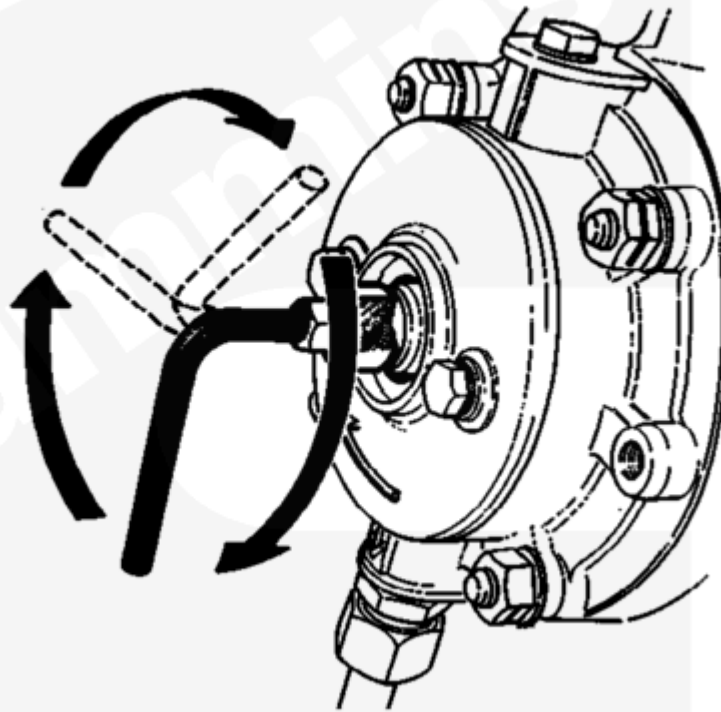
⚠ CAUTION ⚠

The warranty covering the equipment will be invalid in the event of damage caused by unspecified working practices and/or failure to observe the correct specified procedures.

In addition, any relevant classification society or local authority regulations **must** be followed.

Operation

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Figure 1

Start-up Checks. Normal Operation

The following describes the checking procedures for the initial start-up.

Check the function of the hydraulic motor. Make a note of the number of minutes it takes to do one revolution (Figure 1).

The following times are given as a guide for a normal operating condition:

Engine at idle: 10 minutes maximum

Engine at max RPM: 5 minutes maximum.

Note : If the operating indicator does not rotate, or the time for one revolution is more than indicated above, check the pressure drop and make a note of the value. Under normal operating conditions, the value must not exceed 12 psi.

Check for any external oil leaks. If there is an oil leakage, tighten the bolts or change the gaskets as necessary.

When correctly installed and initial start-up performed, the Eliminator™ Filter requires very little attention. However, to provide continued reliable operation, the Eliminator filter hydraulic motor **must** be checked at regular intervals. See Figure 2 below.

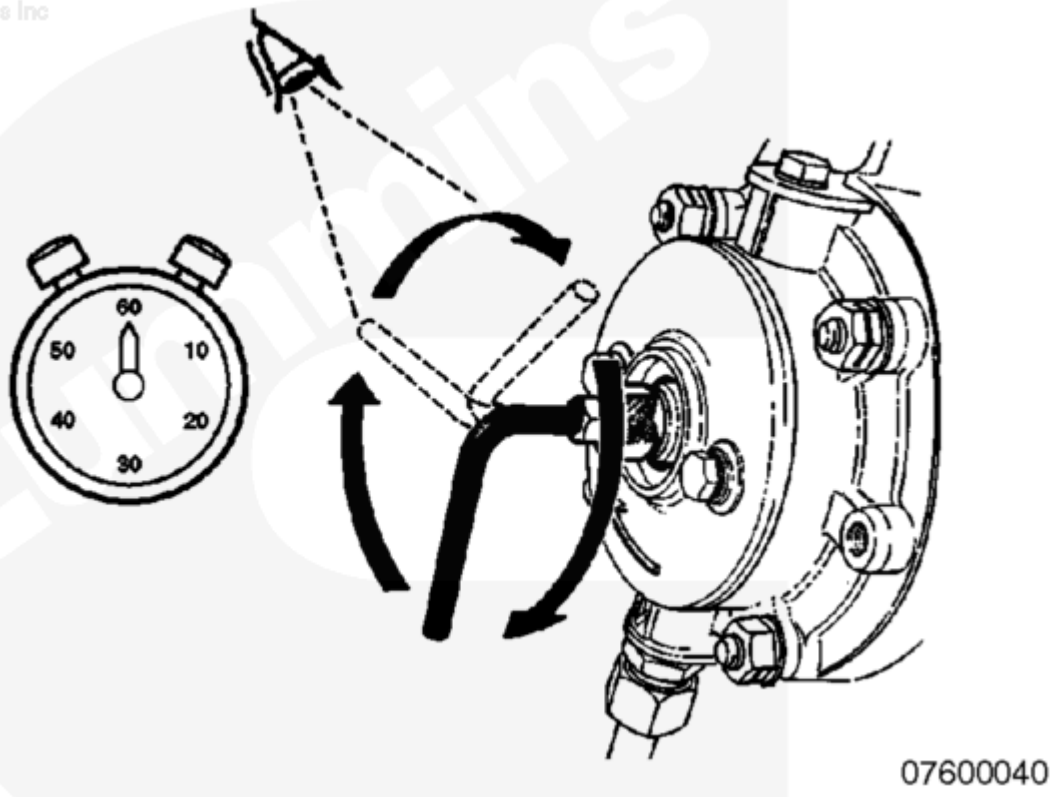


Figure 2

Determination of reference values

Approximately 100 operating hours after the initial start-up or major maintenance procedure, reference values for the pressure drop and rotation of the operating indicator **must** be noted and recorded.

Check the pressure drop and mark this value - Base Line Pressure Drop from 1 to 2 in figure 3.

Note oil temperature and engine speed when measuring the pressure drop.

Check the number of minutes for one revolution of the operating indicator and mark this value - Delta Time (Figure 2).

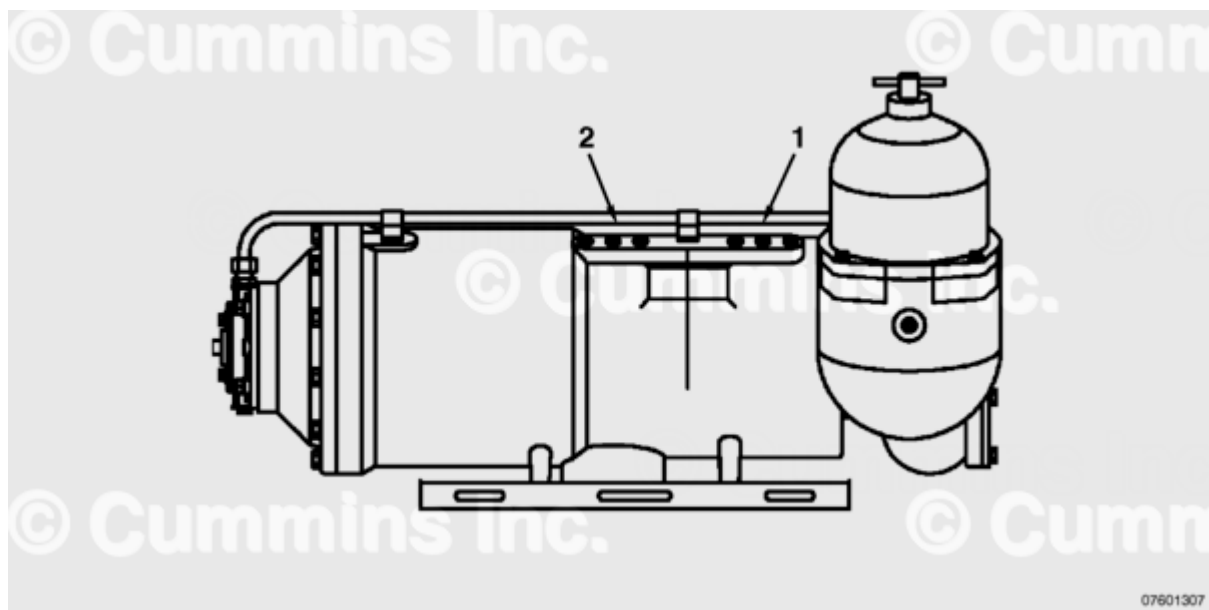


Figure 3

Daily Checks

Every day the pressure drop and operating indicator **must** be checked visually to make sure:

- The pressure drop has **not** increased by more than 2.9 psi above the Base Line Pressure Drop recorded at start up. For example, if Baseline Pressure Drop is 4 psi, maximum pressure drop at the same conditions **must not** exceed 6.9 psi. (Figure 3)
- The operating indicator rotates in a **clockwise** manner. (Figure 1)
- The number of minutes for one revolution is **not** more than 1 minute above the delta time. (Figure 2)
- If pressure drop and operating indicator values are higher than stated, go to Troubleshooting.

Note : These checks **must** be made with the engine at idle.

Troubleshooting Procedures and Techniques

A thorough analysis of the customer's complaint is the key to successful troubleshooting. The more information known about a complaint, the faster and easier the problem can be solved.

The Troubleshooting Symptom Charts are organized so that a problem can be located and corrected by doing the easiest and most logical things first. Complete all steps in the sequence shown from top to bottom.

It is **not** possible to include all the solutions to problems that can occur; however, these charts are designed to stimulate a thought process that will lead to the cause and correction of the problem.

Follow these basic troubleshooting steps:

- Get all the facts concerning the complaint

- Analyze the problem thoroughly
- Relate the symptoms to the basic engine systems and components
- Consider any recent maintenance or repair action that can relate to the complaint
- Double-check before beginning any disassembly
- Solve the problem by using the symptom charts and doing the easiest things first
- Determine the cause of the problem and make a thorough repair
- After repairs have been made, operate the engine to make sure the cause of the complaint has been corrected.

Problems at Initial Start-up. No revolution or slow revolution of the pressure drop indicator

Note : Slow revolution means:

More than 10 minutes for one revolution at idle

More than 5 minutes for one revolution with the engine at max rpm.

Troubleshooting Symptoms Charts

Use the charts on the following pages of this section to aid in diagnosing specific engine symptoms. Read each row of blocks from top to bottom. Follow through the chart to identify the corrective action.

High Pressure Drop Across the Filter		
This is symptom tree t094.		
Cause		Correction
Rough weather stirs up the contaminants from the bottom of the sump (ship only)		Verify that the centrifuge part of the filter is not full of dirt (that it rotates correctly).
OK		
Centrifuge is out of operation		Verify that the centrifuge part of the filter is not full of dirt (that it rotates correctly).
OK		
Engine problems (like piston seizure) occur suddenly		Check the condition of the engine.
OK		

High Pressure Drop Across the Filter		
This is symptom tree t094.		
Cause		Correction
Large amount of new oil has been added to the system, which has absorbed contaminants from the sump walls		Carry out the turn action. Go to Problems at Initial Start-up.

Excessive Vibrations of the Centrifuge		
This is symptom tree t081-002.		
Cause		Correction
Unbalanced due to incorrectly mounted paper insert		Mount the paper insert correctly.
OK		
Unbalanced due to sludge build-up in the rotor		Dismantle and clean the centrifuge.
OK		
Incorrectly tightened rotor cover nut		Tighten the rotor cover nut. Check the polar position of the rotor cover.
OK		
Incorrectly mounted rotor cover due to broken or missing guide pin		Replace centrifuge rotor assembly.

No Rotation of the Centrifuge		
This is symptom tree t108-6.		
Cause		Correction
Missing or damaged o-ring in the centrifuge		Replace the o-ring.
OK		
Incorrectly tightened rotor cover nut		Tighten the rotor cover nut. Check the polar position of the rotor cover.
OK		

No Rotation of the Centrifuge		
This is symptom tree t108-6.		
Cause		Correction
Clogged spindle channel		Dismantle the centrifuge rotor assembly and clean the spindle.
OK		
Clogged nozzles		Clean the nozzles.
OK		
Oil temperature too low		Wait for correct operating temperature.

Leakage Around the Filter		
This is symptom tree t103-3.		
Cause		Correction
Damaged o-ring on filter head		Replace the o-ring (40).
OK		
Damaged o-rings on pressure plug		Replace the o-ring (38).
OK		
Damaged o-rings on drain flange		Replace the o-ring (34), (41).
OK		
Swivel couplings on the return from the hydraulic motor are loose		Tighten the couplings (46) and (48).

Pressure Drop Value Above 40 kPa [5.8 psi]		
This is symptom tree t109-4.		
Cause		Correction
Lubricating oil temperature too low		Operate the engine until oil is at operating temperature.

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No Revolution or Slow Revolution of the Pressure Drop Indicator		
This is symptom tree t108-7.		
Cause		Correction
Lubricating oil temperature too low		Operate the engine until oil is at operating temperature.
OK		
Hydraulic motor is not working properly		Replace the hydraulic motor.

Excessive Noise		
This is symptom tree t081-001.		
Cause		Correction
Rotating parts touching stationary parts		Dismantle the centrifuge. Change the damaged parts as necessary.

Eliminator™ Filter Maintenance Schedule

For scheduled maintenance, go to specific engine Operations and Maintenance Manual.

At end of life or rebuild the entire Eliminator assembly **must** be replaced for a new or reconditioned part.

The below instructions and images are representative of the KV engines, some of the part numbers and quantities will be different for other engine models.

Eliminator™ Filter Centrifuge

A new Cummins Fleetguard™ one-piece Eliminator™ centrifugal oil filter with SpiraTec™ technology has been released. This new unit is disposable. It is intended to reduce the time taken to clean/replace the cone filters. It is preferred that a disposable centrifuge is used in place of the conventional cone stack. Two different disposable centrifuge cartridges are available in the centrifuge parts list. These part numbers are **not** cross compatible.

Cone Stac™ Centrifuge

Tools and Spare Parts Required.

The following are required for cleaning the centrifuge part of the filter:

- Set of spare parts for centrifuge
- Set of inch spanners
- Wooden base/bench
- Mallet
- Pliers
- Silicone grease (for lubrication of gaskets and o-rings)
- Cleaning tray
- Cleaning solvent
- Air/water gun
- Compressed air
- Protective clothing
- Gloves
- Soft brush
- Bucket of diesel fuel or mineral spirits.

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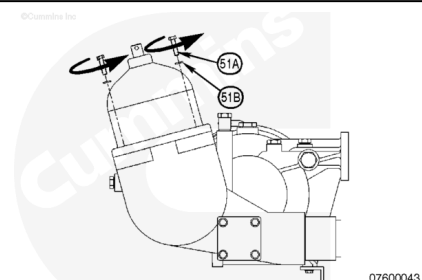
Note : K38, K50, QSK38, and QSK50 Eliminator™ filters utilize a mix of standard (English) and metric thread connectors. QSK45, QSK60, and QSK78 Eliminator™ filters utilize metric thread connectors.

Note : All graphics shown below are of K38 and K50 Eliminator™ filter design; QSK45, QSK60, and QSK78 Eliminator™ filter design is similar.

Remove

⚠ WARNING ⚠

To prevent personal injury make sure that the engine and the rotation of the bowl is completely stopped before any disassembly. Wait at least 5 to 10 minutes after stopping the engine before disassembling the filter.



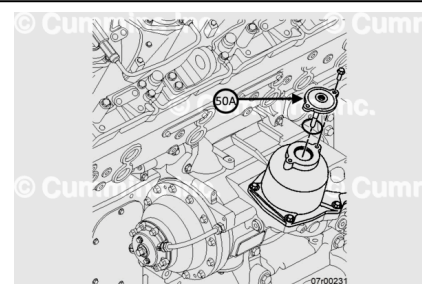
Disassembling the Centrifugal Separator

Remove the four screws (51A) with washers (51B).

Turn the handle counterclockwise, and remove the cover. The handle will stay on the cover.

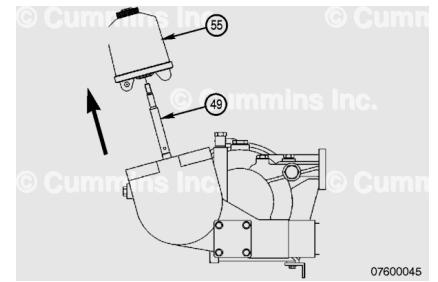
Later models will use a shaft support cover fixed with two capscrews (50A) in place of the handle. Remove the two capscrews and the shaft support cover. Remove the centrifuge cover.

Remove and discard the shaft support cover o-ring seal.



⚠ CAUTION ⚠

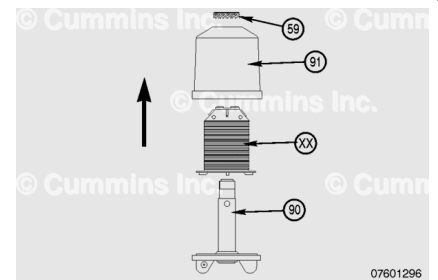
The bowl can be hot. Use gloves to avoid burns and personal injury.



Remove the rotating bowl (55) and place it on a wooden bench.
If Cummins Fleetguard® disposable centrifuge cartridge is being used. Remove disposable centrifuge cartridge.

⚠ CAUTION ⚠

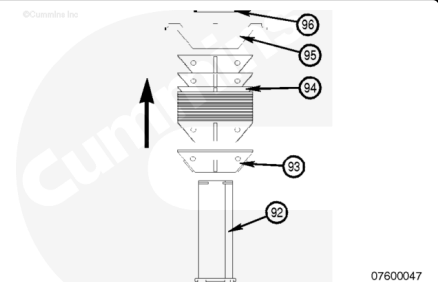
The bowl discs can present sharp edges. Use gloves to avoid personal injury.



Remove the rotor cover nut (59) and rotor cover (91).
Remove the disc stack (xx) from the rotor (90).
Place the disc stack on a wooden base.

⚠ CAUTION ⚠

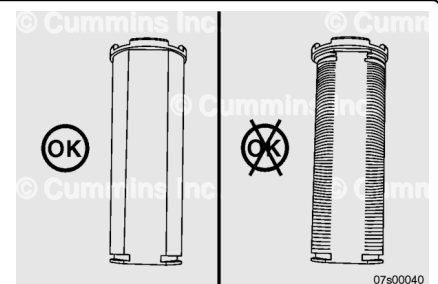
The bowl discs can present sharp edges. Use gloves to avoid personal injury.



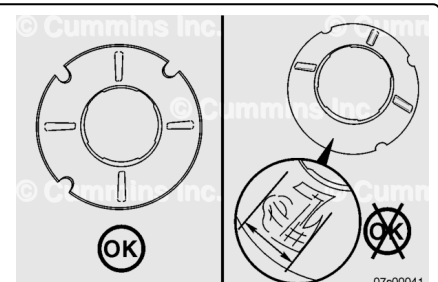
Remove the circlip (96) and distributing cone (95).
Remove the disc (94) from the distributor.
Remove the top disc (93) from the distributor (92).

Clean and Inspect for Re-Use

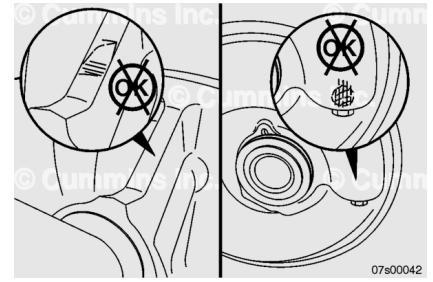
Inspect the disc stack distributor for signs of wear. If there is any wear in the cone stack fastening area, the distributor should **not** be used.



Inspect the cone disc to check that four seat pads are present. If there are not four seat pads present, do not reuse.
Check for wear inside the disc. A small amount of wear is okay.



Inspect the Eliminator™ housing and rotor for signs of contact and wear.



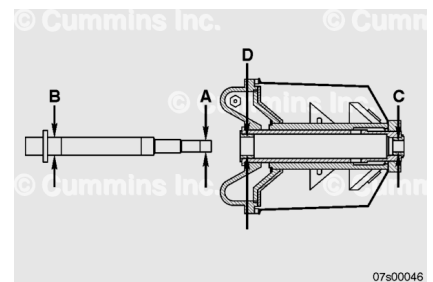
All rotors are correctly balanced before leaving the manufacturer. An out-balance condition can occur if there is an excessive bearing or spindle wear, making necessary to check the bearing tolerances of both parts.

Check the journal bearing clearances as follows:

Maximum Admitted Clearances

- Top (C-A): 0.25 mm
- Bottom (D-B): 0.20 mm

If out of maximum admitted clearance, replace the complete rotor assembly and/or the spindle.

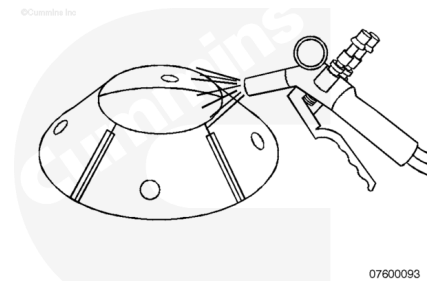


⚠ CAUTION ⚠

If sludge is left, this can create unbalance when the filter is started again.

⚠ CAUTION ⚠

The bowl discs can present sharp edges. Use gloves to avoid personal injury.



Cleaning and Inspecting the Centrifugal Separator

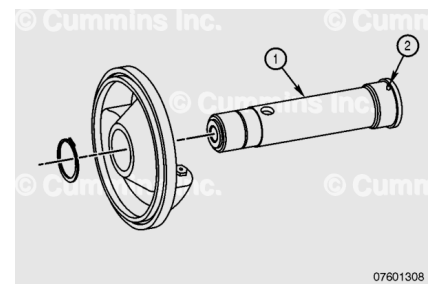
Inspect the inside of the rotor cover. Remove the sludge from the wall without damaging the surface. If a paper insert is used, the sludge is removed by pulling it out. Check that all the sludge is removed. Install a new sludge collecting paper insert.

If necessary put all components into a cleaning tray filled with diesel fuel or mineral spirits.

Remove the last solids retained on the components using compressed air or the air/water gun.

Excessive sludge or improper alignment can cause the bushing (1) to become out of round.

Inspect the alignment pin (2) for wear or damage.



Install

⚠ CAUTION ⚠

The bowl discs can present sharp edges. Use gloves to avoid personal injury.

Assembling the Centrifugal Separator

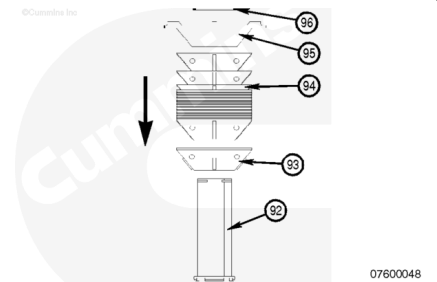
Place the distributor (92) on a work bench.

Install the top disc (93) on the distributor.

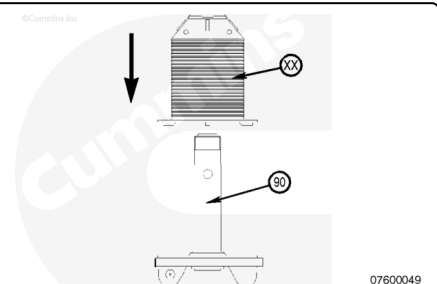
Install the discs (94) on the distributor.

Assemble the distributing cone (95). The correct number of discs can vary. The correct number of discs is achieved when the distributing cone is positioned with the circlip groove on the distributor completely open. Add or remove bowl discs to achieve correct stack height.

Secure the assembly with the circlip (96).



Install the disc stack (xx) on the rotor shaft (90).

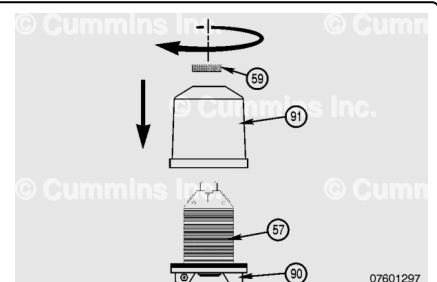


Lubricate the o-ring. Install a new o-ring (57) on the rotor (90).

Install a new sludge collecting paper insert into the rotor cover (91).

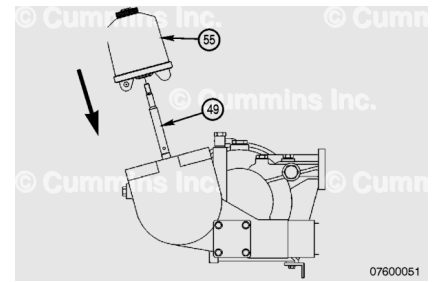
Install the rotor cover (91).

Hand tighten the nut (59).



Mount the rotating bowl (55) on the spindle (49).

If Cummins Fleetguard® disposable centrifuge cartridge is being used. Mount disposable centrifuge cartridge on the spindle.

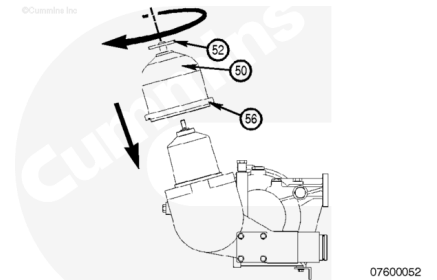


Install a new o-ring (56) on the cover (50).

Lubricate the o-ring.

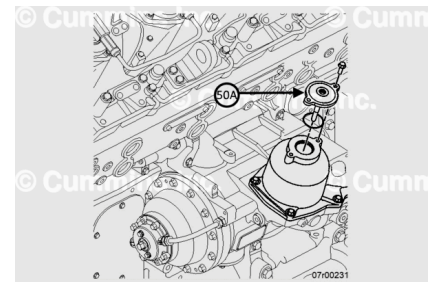
Install the cover (50) on the filter, and turn the handle (52) clockwise.

Torque Value: 25 n•m [221 in-lb]



Later models will use a shaft support cover fixed with two capscrews (50A) in place of the handle. Install the two capscrews and tighten.

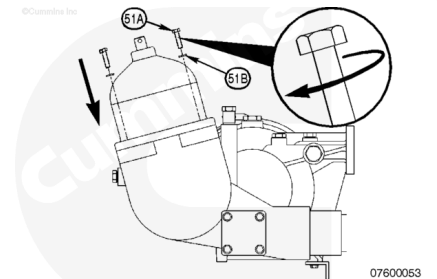
Torque Value: 25 n•m [221 in-lb]



Install the four capscrews (51A) and washers (51B).

Tighten the screws.

Torque Value: 25 n•m [221 in-lb]



Hydraulic Motor

Tools and Spare Parts Required

The following is required to overhaul the hydraulic motor:

Note : The hydraulic motor is no longer a serviceable component and must be replaced with a new or reconditioned part purchased from a parts distribution center (PDC).

Tools and Spare Parts Required

- Set of gaskets for the hydraulic motor (see Parts listings)
- Two M10 screws

- Wooden base/bench
- Mallet
- 2 mm [0.08 in] and 3 mm [0.12 in] pin punch
- Pliers
- Silicone grease
- Cleaning tray
- Compressed air
- Soft brush
- Bucket of diesel fuel or mineral spirits
- Emery cloth (fine grade).

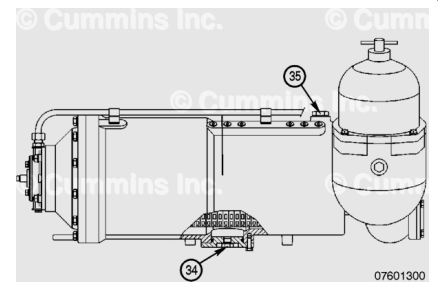
Removing the Hydraulic Motor

⚠ CAUTION ⚠

To prevent personal injury make sure that the engine is stopped before any dismantling.

⚠ CAUTION ⚠

The oil can be hot. Use gloves to prevent burns or personal injury.



Place a suitable container under the drain plug (34).

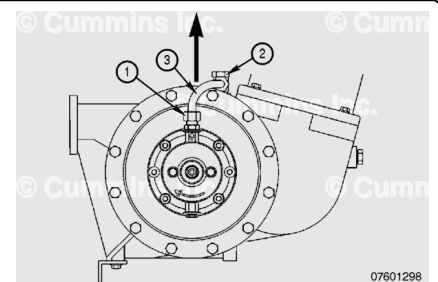
Remove the plug (34) and let the oil contained in the filter drain.

Note : Removing the vent plug (35) helps to empty the filter.

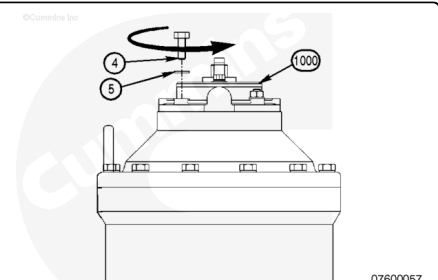
Install the drain plug (34) and vent plug (35) in the filter body.

Remove the coupling (1) and the banjo screw (2) on the hydraulic motor return pipe.

Remove the pipe supports and the return pipe (3).



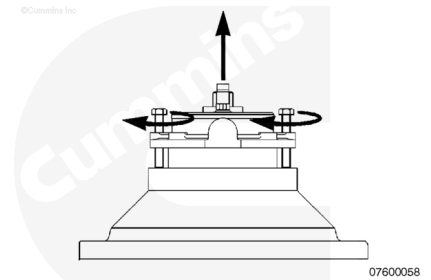
Remove the four screws (4) and washers (5) from the hydraulic motor (1000).



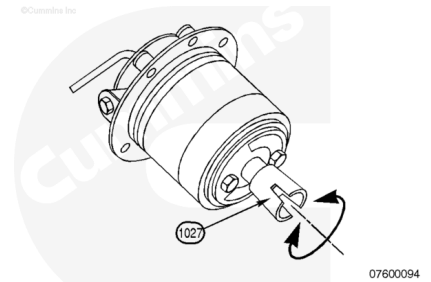
Install two M10 screws on opposite sides of the hydraulic motor and tighten them alternately half a turn.

When the hydraulic motor is raised out of the filter head, lift it out.

Remove the two M10 screws. Discard the Motor O-ring (39).



Rotate the driver (1027) by hand. If it can be rotated in **only** one direction, the teeth of the ratchet are OK. If it can be rotated in both directions, the motor **must** be replaced.



Installing the Hydraulic Motor

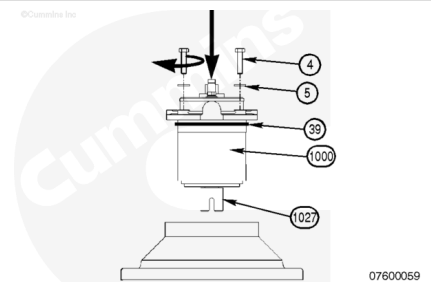
Install a new o-ring (39) on the hydraulic motor (1000), and lubricate it with engine oil.

Install the hydraulic motor back into the filter head.

Note : Make sure the driver (1027) is in the correct position, corresponding to the driving pin of the distributor.

Install and tighten the four capscrews (4) and washers (5) in the hydraulic motor.

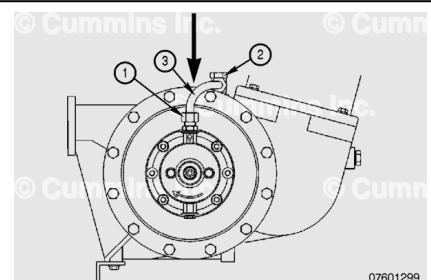
Torque Value: 25 n·m [221 in-lb]



Install the hydraulic motor return pipe (3).

Tighten the banjo screw (2) and coupling (1) on the hydraulic motor pipe.

Torque Value: 40 n·m [30 ft-lb]



Filter Elements

Note : This cleaning and inspecting procedure is **only** recommended in the case of a malfunction event or if in service cleaning is required.

Note : Filter elements are no longer purchased in pairs, all elements must be replaced as an assembly.

Cleaning and Inspecting the Filter

⚠ CAUTION ⚠

To prevent personal injury make sure that the engine is stopped before performing any disassembly.

⚠ CAUTION ⚠

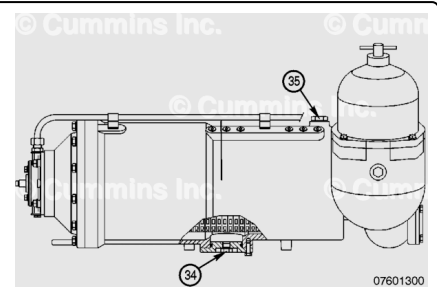
The oil may be hot. Use gloves to avoid burns.

Removing the Filter

Place a suitable container under plug (34). Remove the plug (34) to allow the oil in the filter to drain.

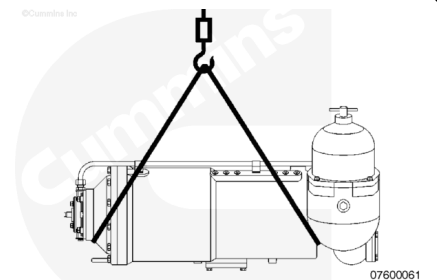
Note : Removing the vent plug (35) helps to empty the filter.

Install the drain plug (34) and vent plug (35) in the filter body.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



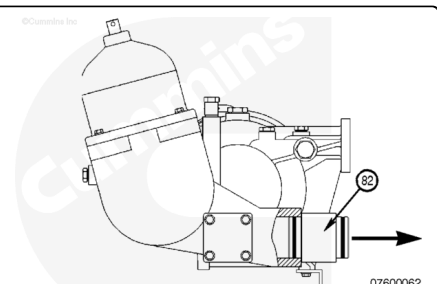
Remove the two capscrews from the lower support.

Remove four capscrews and install the four guide studs.

Install two straps to support the filter. Remove the eight capscrews holding the filter to the engine. Remove the filter from the engine. Place the filter vertically on a work bench.

Remove the gaskets.

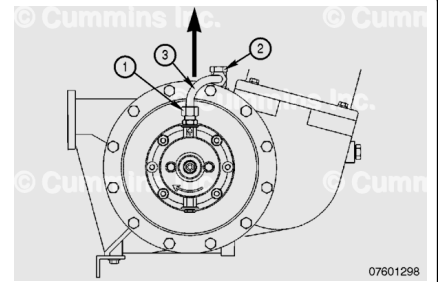
Remove the connection tube (82).



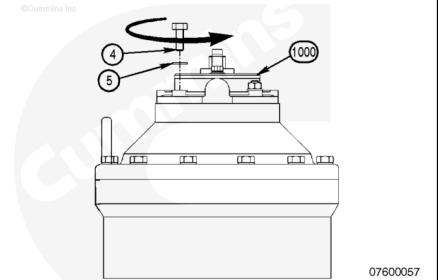
Disassembling the Filter

Loosen the banjo screw (2) and coupling (1) on the hydraulic motor return pipe.

Remove the return pipe (3).



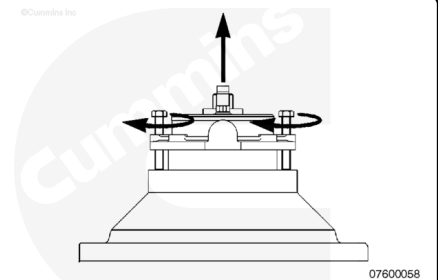
Remove the four screws (4) and washers (5) from the hydraulic motor (1000).



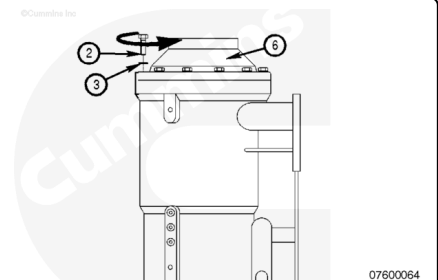
Install two M10 screws on opposite sides of the hydraulic motor, and tighten them alternately half a turn until the hydraulic motor is raised out of the filter head.

Remove the hydraulic motor. Discard the Motor O-ring (39).

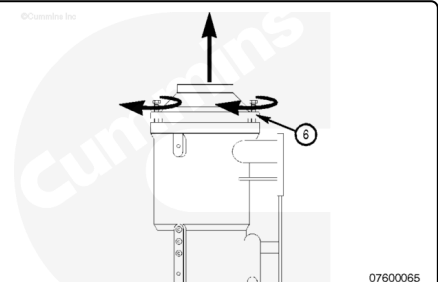
Remove the two M10 screws.



Remove the twelve screws (2) and washers (3) from the filter head (6).



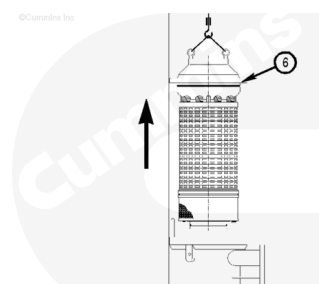
Install two 1/2-13 UNC screws in opposite holes on the filter head (6). Tighten them alternately half a turn until the filter head separates from the housing.



This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Install two 1/2-13 UNC eye bolts in opposite holes on the filter head (6).

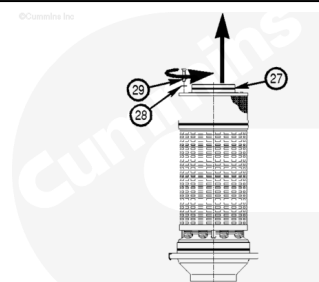
Remove the filtering unit and place it on a work bench.



07600066

Disassembling the Filtering Unit

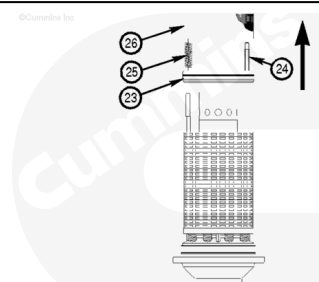
Remove the four screws (29) and washers (28) from the cover (27). Remove the cover.



07600067

Remove the full flow cover (23), with the studs (24), strainer (26), and springs (25).

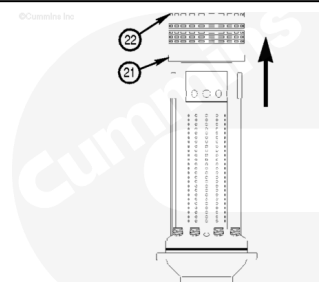
Note : Springs (25) are required only for Eliminator™ Filters manufactured before September 2016.



07600068

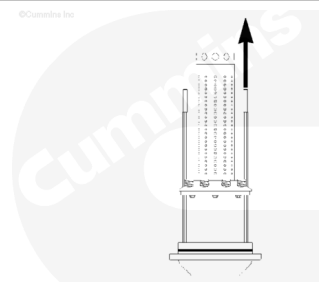
Remove the full flow filtering elements (22) one by one.
Remove the companion spacer (21).

Note : Two filter elements are available and are not allowed to be mixed. The recommended element to use is part number 3651716 however, some older eliminators may be found to use part number 3393557.



07600069

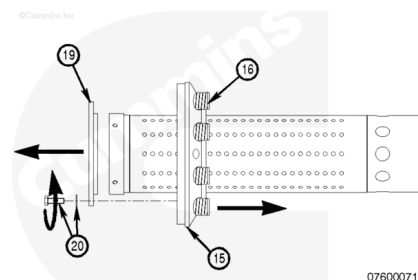
Remove the sleeve/distributor assembly and place it on a wooden bench.



07600070

Remove the four screws (20) and washers from the cover (19).
Remove the cover and spacer (15).

Note : The springs (16) are kept in place by screws in the spacer. Do not remove the springs.

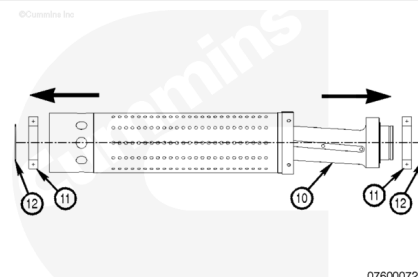


Remove the circlips (12) and ball bearings (11).

Inspect the ball bearings to make sure they rotate freely.

Note : Replace the bearings if seized or worn.

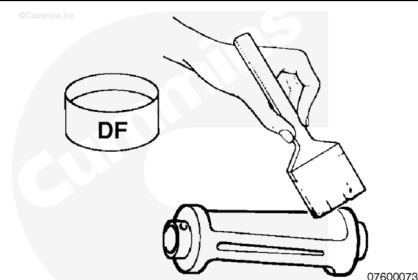
Remove the distributor (10) and place it on a wooden bench.



Cleaning and Inspection of the Filtering Unit

Note : Perform the following on all the components except the filter elements.

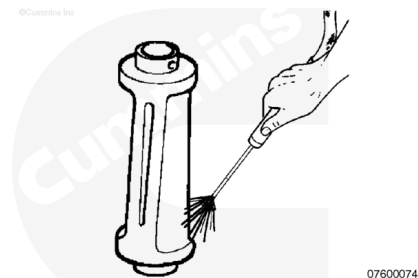
Carefully clean all the components (except the filter elements) with clean diesel fuel or mineral spirits and a soft brush.



⚠ WARNING ⚠

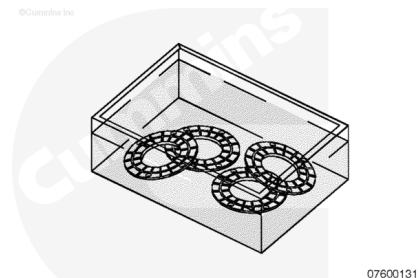
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove the last solids retained on the components using compressed air.



⚠ WARNING ⚠

Follow the manufacturers instructions carefully when using solvents. Wear protective clothing and goggles. Use only in well ventilated spaces. Keep all flames, sparks, arcing switches and equipment, and pilot lights out of work area and areas sharing ventilation. Do not smoke. Solvent can be flammable. If the solvent comes into contact with your eyes, rinse thoroughly with cold water and seek immediate medical attention.



⚠ CAUTION ⚠

Do not use acid-based cleaning solvents. They can damage the filter elements.

Note : Keep one full-flow filter element as a guide for the reassembly of the filter halves after cleaning.

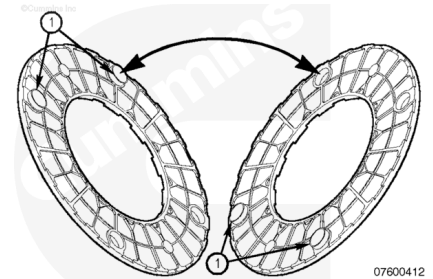
Note : The solvent must be new or filtered to 5 microns for best results.

Cleaning the Filter Elements

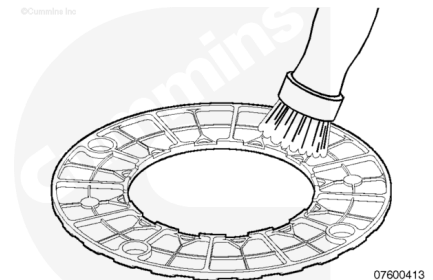
Put the filter elements into a cleaning tray filled with cleaning solvent. Soak the filter elements in the tray at least 15 minutes.

Remove one of the element from the cleaning tray and separate it into halves.

Note the orientation of the element halves. The sides with the dowel rings (1) are the clean sides of the screens.

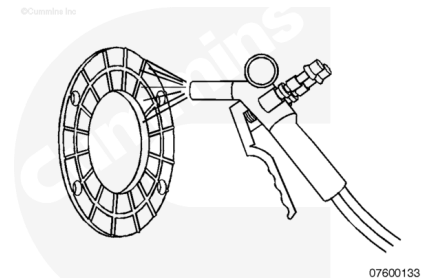


Brush the element halves with a soft (silk) bristle brush.



⚠ WARNING ⚠

When using an air-water gun, wear safety glasses and/or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ CAUTION ⚠

To prevent damage to the screens, air pressure must not exceed 7 bar [102 psi]. Water pressure must not exceed 5 bar [73 psi]. Water must be hot to remove solvent and dirt effectively.

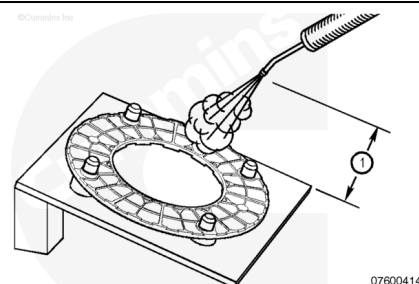
Clean the elements with one of the following two methods:

⚠ WARNING ⚠

When using an air-water gun, wear safety glasses and/or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

Maximum pressure is 20 bar [290 psi]. Maximum temperature is 100°C [212°F]. Minimum distance between the nozzle and the element is 20 cm [7.9 in] (1). Do not start the high-pressure cleaner with the element in front of the nozzle. Do not use any cleaning chemicals in the high-pressure cleaner.

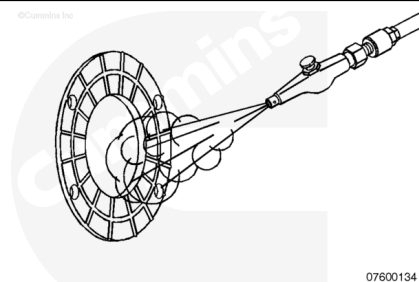


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Elements **must** be rinsed thoroughly with hot water to remove any residual solvent.

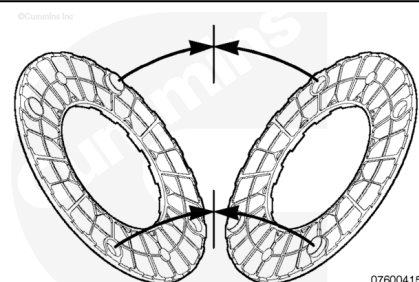
Carefully inspect the filter element halves for any damage to the mesh and screen filter body and any dirt. If any elements are found to be damaged, all elements in the stack **must** be replaced. **Always** check the filter screen for cracks or leak paths. If the elements are still dirty, repeat the cleaning procedure. Increase the soak time from 15 minutes to 30 minutes. The element is clean when light can be seen easily through the inner mesh.



Assemble the element halves and repeat the cleaning steps on the next element from the tray.

When all the elements have been satisfactorily cleaned and assembled according to the guide element, clean the guide element.

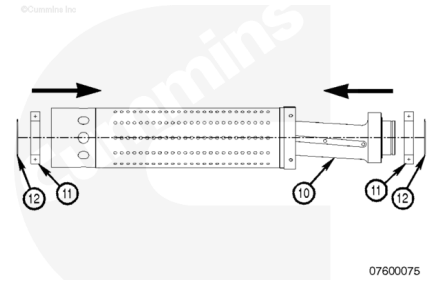
Note the orientation of the element halves. The sides with the dowel rings are the clean sides of the screens.



Assembling the Filtering Unit

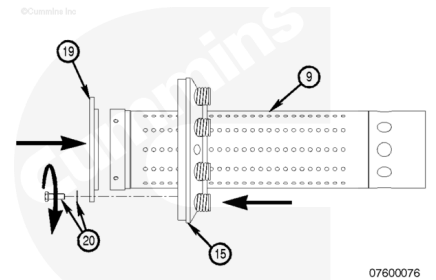
Use clean engine oil to lubricate the distributor (10). Install it back into the sleeve.

Assemble the ball bearings (11) and circlips (12) on the distributor.



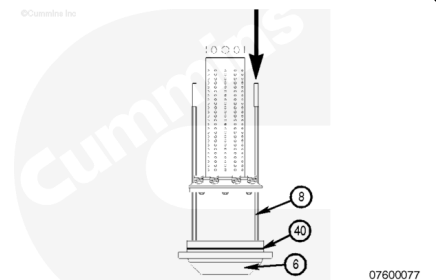
Assemble the spacer (15) on the sleeve (9). Make sure that the four locating pins are in the correct position. Install the cover (19) and four capscrews and washers (20).

Torque Value: 35 N·m [26 ft-lb]



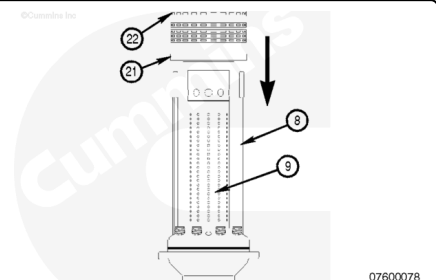
Install a new o-ring (40) on the filter head (6). Lubricate the o-ring with silicone grease.

Install the sleeve/distributor assembly on the filter head (6) with rods (8).



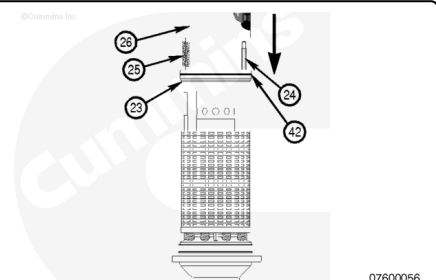
Install the assembly vertically on a work bench.

Assemble the companion spacer (21) and the clean full-flow filtering elements (22) one by one on the full-flow rods (8). Make sure the same number of elements are installed that were removed.



Assemble the full-flow cover (23), strainer (26) and springs (25). Use clean engine oil and lubricate and install a new o-ring (42) in the full-flow cover (23).

Note : Springs (25) are required only for Eliminator™ Filters manufactured before September 2016.



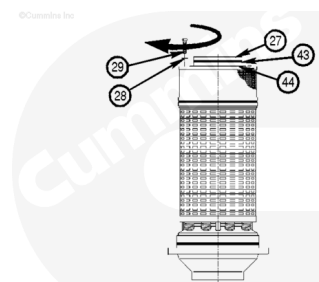
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Use clean engine oil and lubricate and install new o-rings (43) and (44), in the cover (27).

Secure the cover (27) to the rods, using four screws (29) and washers (28). Install a new o-ring (43) in the cover (27) and lubricate it.

Torque Value: 25 n·m [221 in-lb]

Turn the filtering unit upside-down.



07600079

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Assembling the Filter

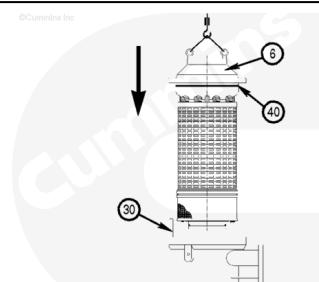
Lubricate and install a new o-ring (40) in the filter head (6).

Install two 1/2-13 UNC eye bolts on opposite sides of the filter head (6).

Install a strap between the eye bolts.

Install the filtering unit carefully into the filter body.

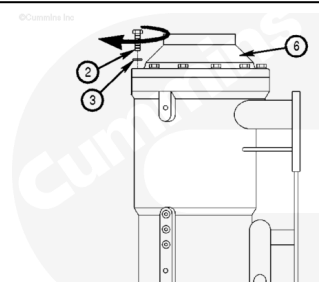
Make sure the holes in the filter head are facing the guide rods (30) of the filter body.



07600080

Install and tighten the twelve screws (2) and washers (3) into the filter head (6).

Torque Value: 35 n·m [26 ft-lb]



07600081

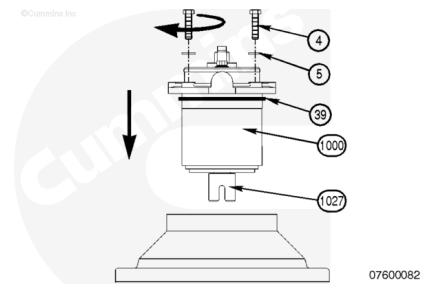
Use clean engine oil to lubricate and install a new o-ring (39) on the hydraulic motor (1000).

Install the hydraulic motor back into the filter head.

Make sure that the driver (1027) is in the corresponding position to the driving pin (**not** shown) of the distributor.

Install and tighten the four screws (4) and washers (5) in the hydraulic motor (1000).

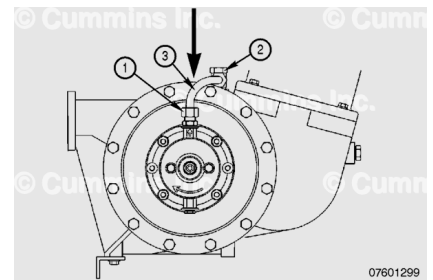
Torque Value: 25 n·m [221 in-lb]



Connect the hydraulic motor return pipe (3).

Install and tighten the banjo screw (2) and coupling (1) on the hydraulic motor pipe.

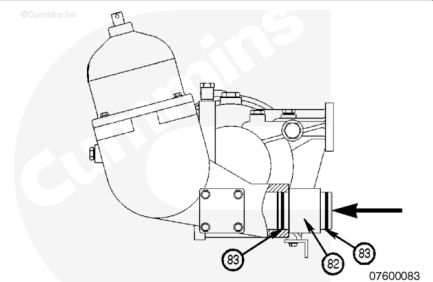
Torque Value: 40 n·m [30 ft-lb]



Mounting the Filter

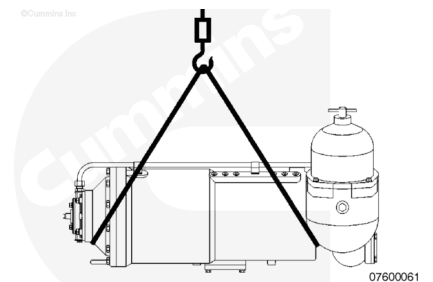
Use clean engine oil to lubricate and install two new o-rings (83) on the connection tube (82).

Install the connection tube to the filter body.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the four guide studs. Install the new gasket over the guide studs.

Install the filter onto the engine.

Do **not** forget to install new gaskets between the filter body and cylinder block.

Install and tighten eight capscrews.

Torque Value: 40 n·m [30 ft-lb]

Remove and replace the four guide studs with the four remaining capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

Eliminator™ Filter (Pre September 2016)

⚠ CAUTION ⚠

To avoid personal injury make sure that the engine is stopped before performing any work.

Cleaning the Filter Elements

This procedure is valid for cleaning the filtering elements when an unexpected clogging appears.

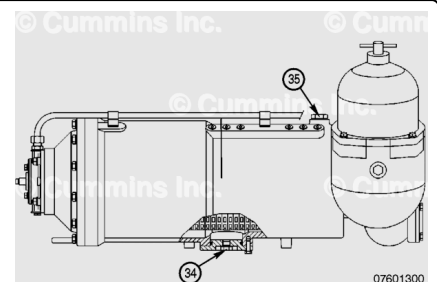
⚠ CAUTION ⚠

The oil can be hot. Use gloves to avoid personal injury.

Place a suitable container under the drain plug (34). Remove the plug and let the oil drain.

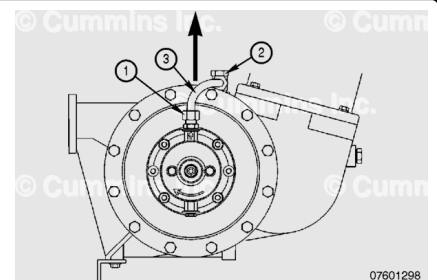
Note : Remove the plug (35) to help drain the filter.

Install the plugs (34) and (35) in the filter body.

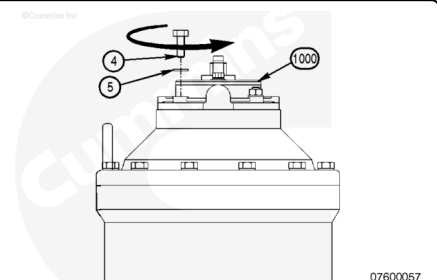


Remove the banjo screw (2) and coupling (1) from the hydraulic motor return pipe.

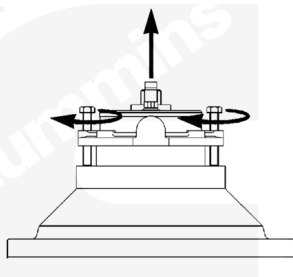
Remove the pipe (3).



Remove the twelve screws (4) and washers (5) from the filter head (1000).

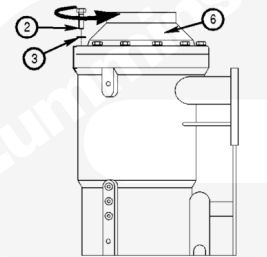


Install two 1/2-13 UNC screws in the opposite holes on the filter head, tighten them alternately half a turn. When the hydraulic motor is raised enough, remove the filter head.



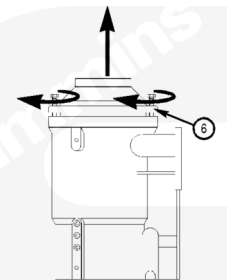
07600058

Remove the twelve screws (2) and washers (3) from the filter head (6).



07600064

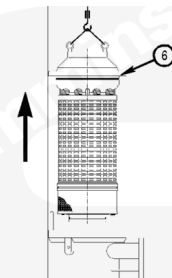
Install two 1/2-13 UNC screws in opposite holes on the filter head (6). Tighten them alternately half a turn.



07600065

⚠ WARNING ⚠

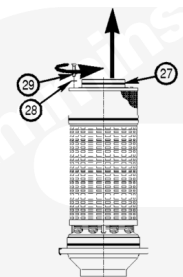
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



07600066

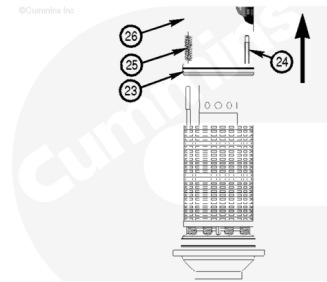
Remove the filtering unit (6) and carefully place it on a work bench.

Remove four screws (29) and washers (28) from the cover (27). Remove the cover.



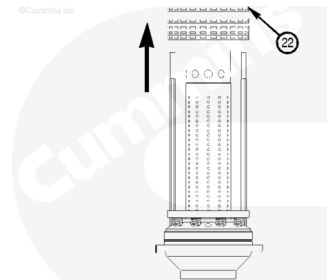
07600067

Remove the full flow cover (23) with studs (24), strainer (26) and springs (25).



07600068

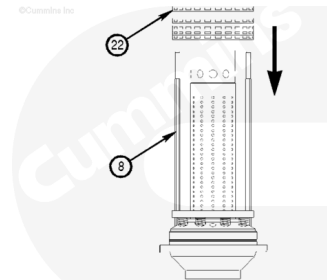
Remove the full flow filtering elements (22) one by one.
Clean the elements.



07600084

Assemble the clean full flow filtering elements (22) one by one on the full flow rods (8).

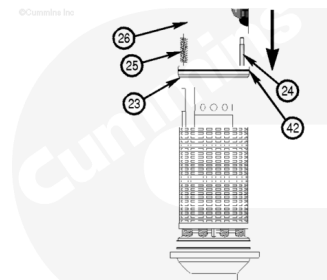
Take care **not** to damage the elements on the rods. Make sure that the same number of elements are installed as were removed.



07600086

Assemble the full flow cover (23) with the strainer (26) and springs (25).

Use clean engine oil to lubricate and insert a new o-ring (42) in the full flow cover (23).

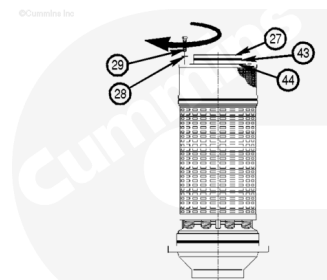


07600056

Install the cover (27) with four capscrews (29) and washers (28).

Use clean engine oil to lubricate and insert new o-rings (43) and (44) in the cover (27).

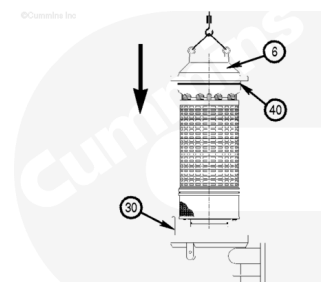
Torque Value: 25 n·m [221 in-lb]



07600079

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



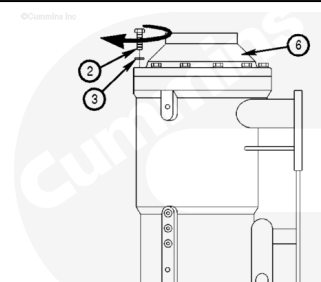
07600080

Carefully install the filtering unit (6) into the filter body.

Make sure that the holes in the filter head (40) are facing the guiding rods (30) in the filter body.

Install and tighten the twelve screws (2) and washers (3) in the filter head (6).

Torque Value: 40 n·m [30 ft-lb]



07600081

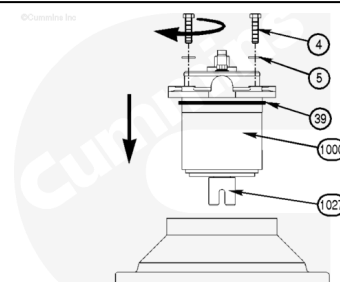
Use clean engine oil to lubricate and install a new o-ring (39) on the hydraulic motor (1000).

Install the hydraulic motor into the filter head.

Make sure the driver (1027) is in the correct position, corresponding to the driving pin (**not** shown) of the distributor.

Install and tighten four capscrews (4) and washers (5) in the hydraulic motor (1000).

Torque Value: 25 n·m [221 in-lb]

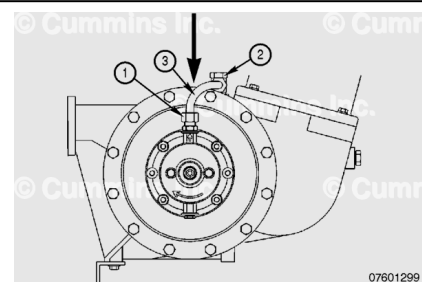


07600082

Install the hydraulic motor return pipe (3).

Install and tighten the banjo screw (2) and coupling (1).

Torque Value: 40 n·m [30 ft-lb]

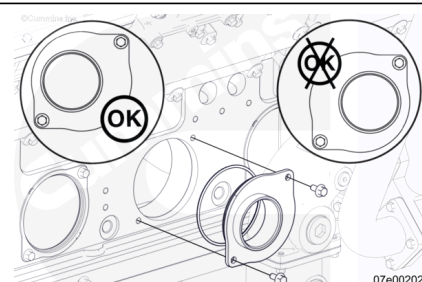


07601299

Three-piece Bracket

⚠ CAUTION ⚠

The Eliminator™ filter can be installed on either side of the engine. Check to be sure the lubricating oil transfer tube has been installed correctly. The filter must be on the opposite



07e00202

bank from the oil transfer tube. If the Eliminator™ filter is installed on the wrong bank, the engine will be severely damaged because of the lack of lubrication.

Install the Eliminator™ filter handhole cover with a new gasket.

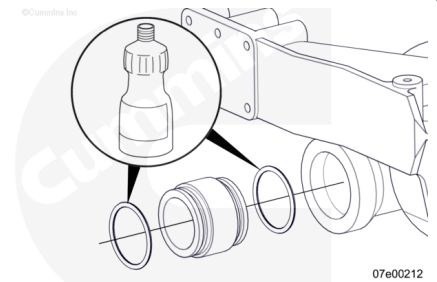
Verify handhole cover is in correct position on engine.

Tighten the capscrews in handhole cover.

Torque Value: 45 n·m [33 ft-lb]

Lubricate the new o-ring seals with vegetable oil and install on both ends of the connection tube.

Install the connection tube into the filter housing.



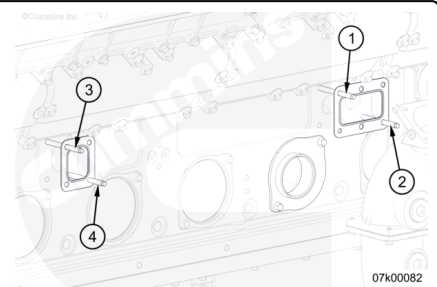
Install four guide studs to the engine in locations illustrated.

Do **not** use threaded guide studs. This will aid in guide stud removal once the Eliminator filter is installed.

Install the gaskets over the guide studs.

Make sure the six mounting holes are visible through each gasket.

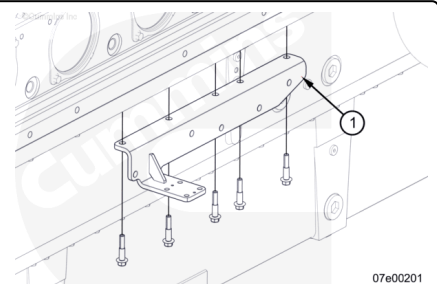
The gasket part number **must** face away from the cylinder block.



Use the mounting capscrews to install the rear bracket (1) to the lubricating oil pan.

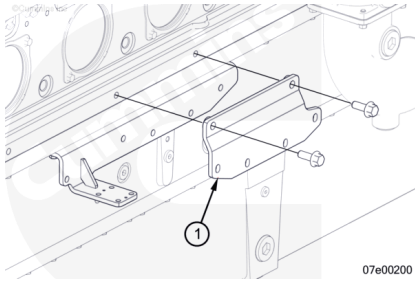
Tighten the capscrews.

Torque Value: 45 n·m [33 ft-lb]

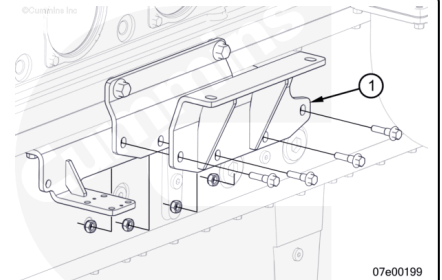


Use the bracket mounting capscrews to install the middle bracket (1). Tighten the capscrews.

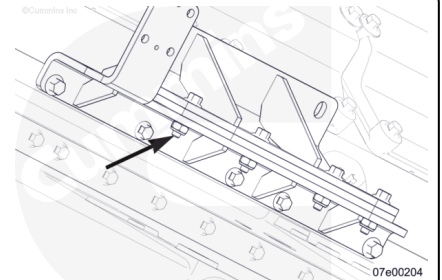
Torque Value: 80 n·m [59 ft-lb]



Use the bracket cap screws and nuts to install the front bracket (1).

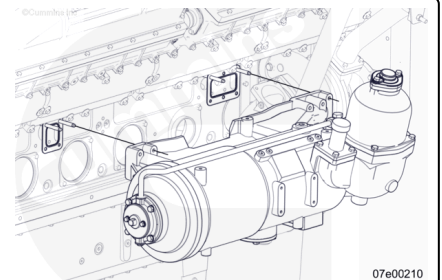


The cap screws should go through all three brackets.
Do **not** tighten the cap screws and nuts.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the filter over the guide studs while guiding the connection tube into the handhole cover.

If the installation requires spacers, install spacer with gaskets between filter housing and the cylinder block.

The front bottom cap screw **must** be inserted before the Eliminator™ filter is moved into place, due to limited clearance.

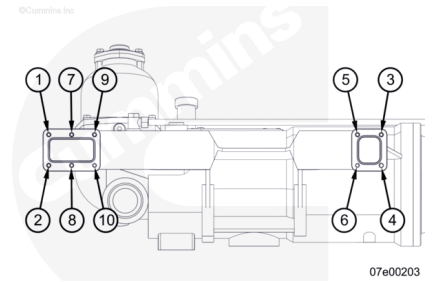
Install six cap screws and washers and hand-tighten.

Remove the guide studs and install the remaining cap screws and washers, hand-tighten to the eliminator manifold.

Tighten the capscrews in sequence illustrated.

Torque Value: 45 n•m [33 ft-lb]

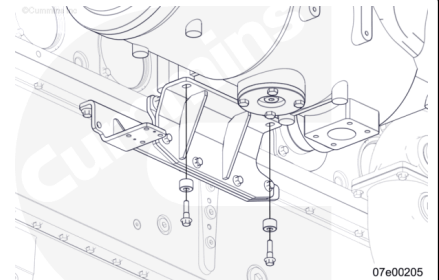
Due to the difficulty in accessing the lower capscrews, the use of a 3/8 drive socket and torque wrench is recommended to gain access to achieve the correct torque value on these capscrews.



Use capscrews and spacers to secure the filter head front support bracket to the bottom of the filter head.

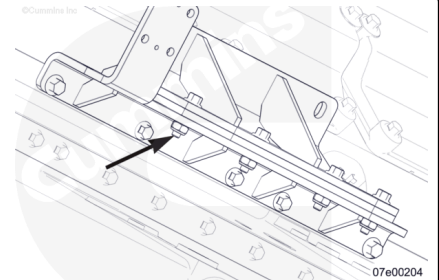
Install and tighten the two lower support capscrews.

Torque Value: 45 n•m [33 ft-lb]



Tighten the capscrews going through all three support brackets.

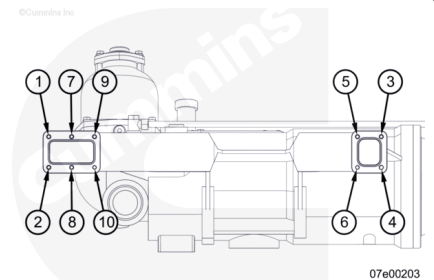
Torque Value: 45 n•m [33 ft-lb]



Loosen capscrews in sequence illustrated by at least one full turn.

Tighten capscrews in sequence illustrated in order from 1 to 10.

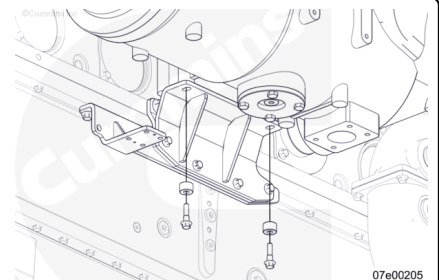
Torque Value: 45 n•m [33 ft-lb]



Loosen capscrews by at least one full turn.

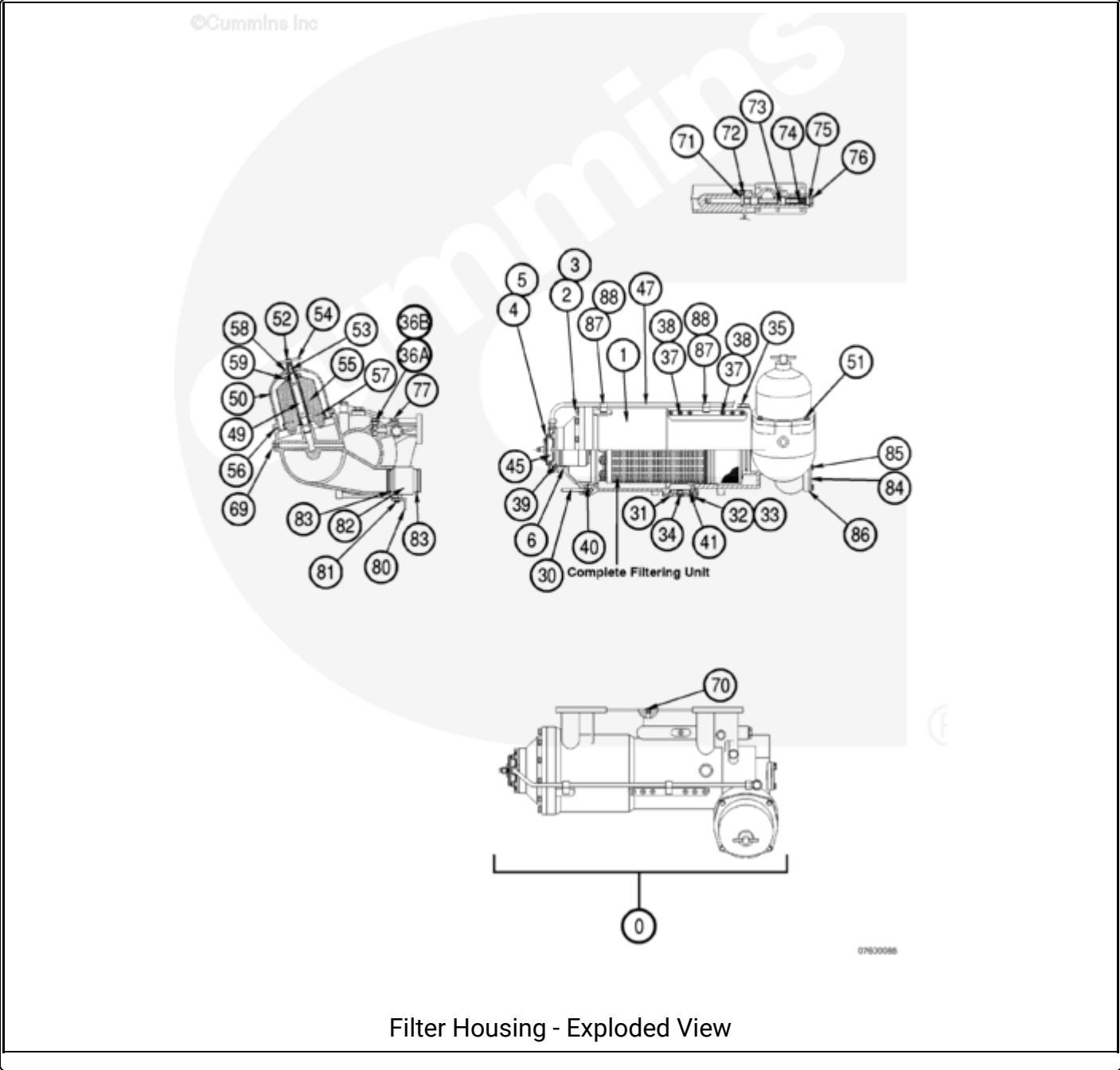
Tighten both capscrews again.

Torque Value: 45 n•m [33 ft-lb]



Parts General Information

This chapter shows the service parts and quantities required for the Eliminator™ Filter. This list is representative of the KV engines, some of the part numbers and quantities will be different for other engine models. Contact a Cummins® Authorized Repair Location for part number information.



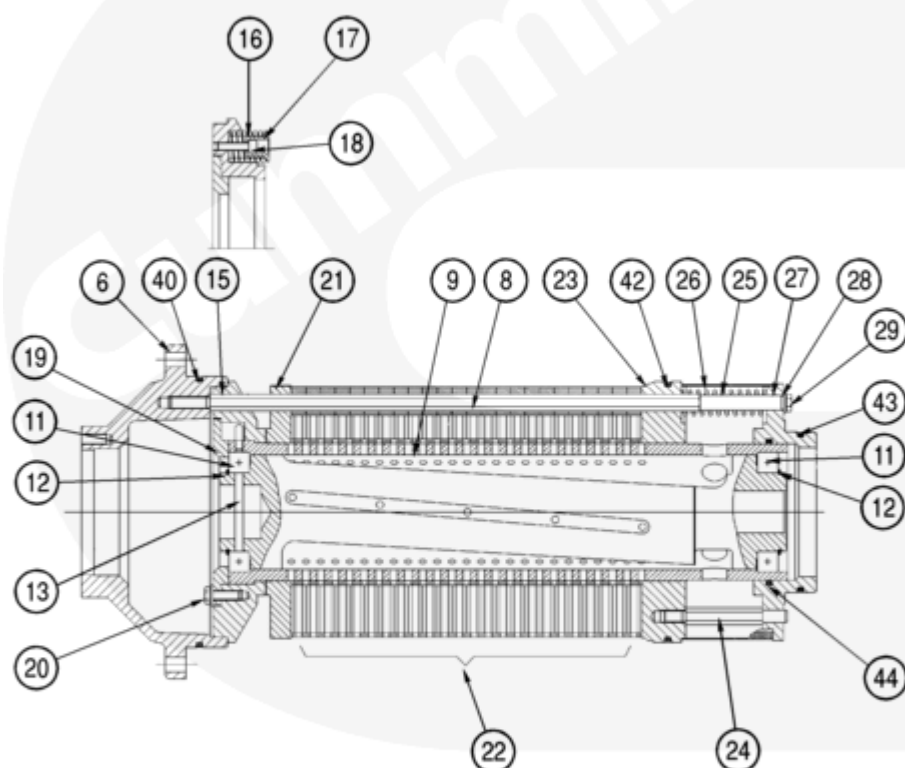
Filter Housing Parts list

Filter Housing			
Item	Description	Part Number	Quantity
1	Filter Body	4007911	1
2	Screw	3065490	12
3	Washer	4016955	12

Filter Housing			
Item	Description	Part Number	Quantity
4	Screw	3906621	4
5	Washer	3633004	4
6	Motor Housing	-	1
30	Locating Pin	-	2
31	Flange	-	1
32	Screw	3065497	4
33	Washer	3633004	4
34	Drain Plug	4332469	1
35	Plug	4332468	1
36A	Nozzle	-	1
36B	O-ring	-	1
37	Plug	4332471	6
38	O-ring	-	6
39	O-ring	3393584	1
40	Filter Head	4007912	1
41	O-ring	3393437	1
45	Hydraulic Motor	6660301	1
47	Hydraulic Motor Return Pipe	3649823	1
49	Centrifuge Shaft Assembly	3649458	1
50	Centrifuge Cover	4016423	1
51	Screw	3906621	4
52	Hollow Screw	-	1
53	Circlip	-	1
54	Pin	-	1
55	Centrifuge	3649571	1
56	O-ring	3393586	1
57	O-ring	3393587	1
58	O-ring	3633001	1
59	Rotor Cover Nut	-	1
69	Plug	3678873	1
70	Plug	3037536	1
71	Stop	-	1
72	Plug	-	1
73	Valve	-	1

Filter Housing			
Item	Description	Part Number	Quantity
74	Spring	-	1
75	Gasket	-	1
76	Plug	-	1
77	Plug	3633039	1
80	Filter Support	-	1
81	Screw	-	2
82	Connecting Pipe	4016425	1
83	O-ring	3633325	2
84	Flange	-	1
85	Screw	-	4
86	O-ring	3631940	1
87	Pipe Support	4086075	2
88	Screw	-	2

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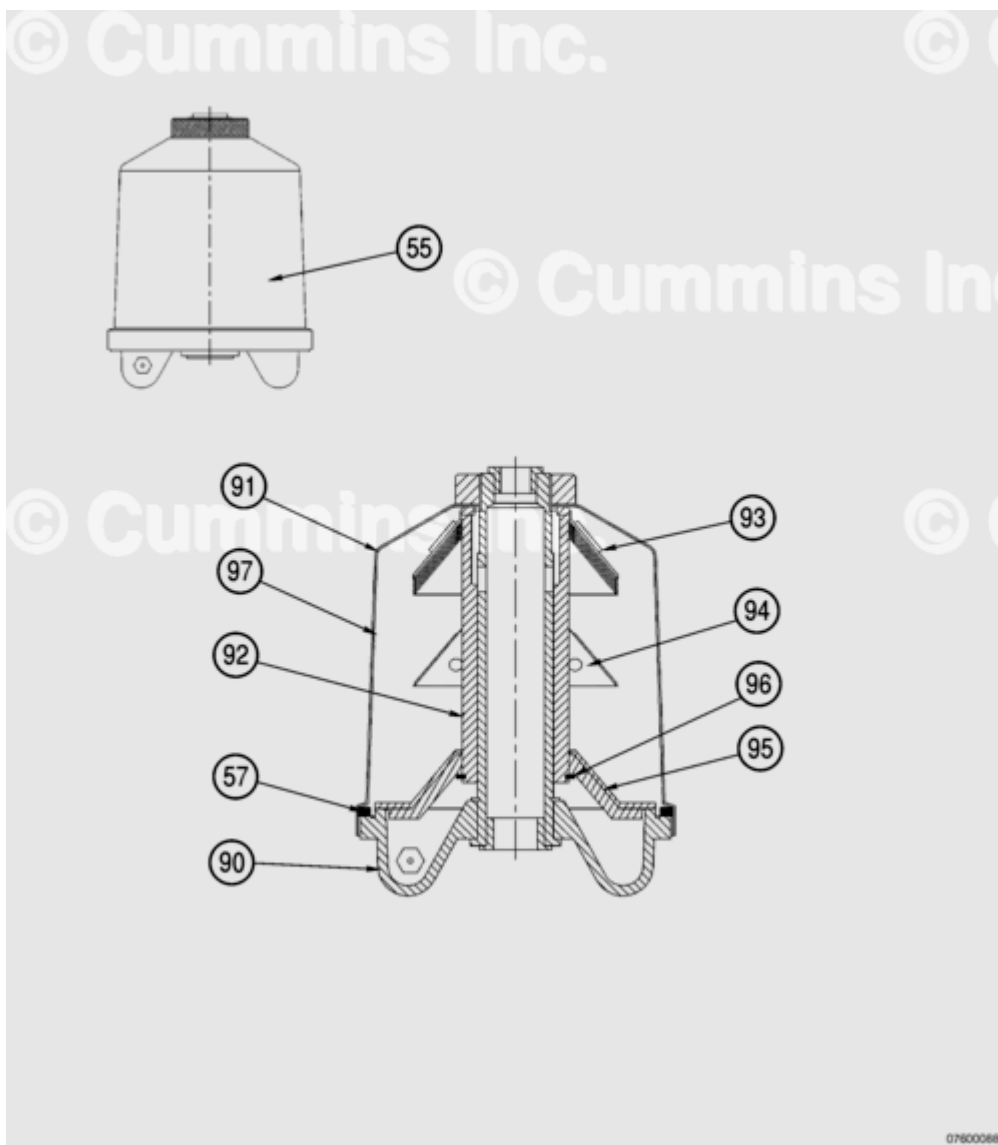
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Filtering Unit - Exploded View

Filtering Unit Parts list

Filtering Unit			
Item	Description	Part Number	Quantity
6	Filter Head	4007912	1
7	Washer	4016955	4
8	Assembling Rod	-	4
9	Sleeve/Distributor Assembly	4007913	1
11	Ball Bearing	3393560	2
12	Circlip	3393561	2

Filtering Unit			
Item	Description	Part Number	Quantity
13	Driving Pin	3393562	1
15	Spacer Assembly	4007914	1
16	Spring	3393564	8
17	Spring Guide	-	8
18	Cover	-	1
19	Cover	-	1
20	Screw	-	4
21	Companion Spacer	-	1
22	Filtering Element	3393557	24
22	Filtering Element	3651716	30
23	Full Flow Cover	-	1
24	Column	-	4
25	Spring	-	4
26	Strainer	3393567	1
27	Cover	-	1
28	Washer	3633004	4
29	Screw	3065497	4
40	O-ring	3393565	1
42	O-ring	3393569	1
43	O-ring	3393570	1
44	O-ring	3393571	1



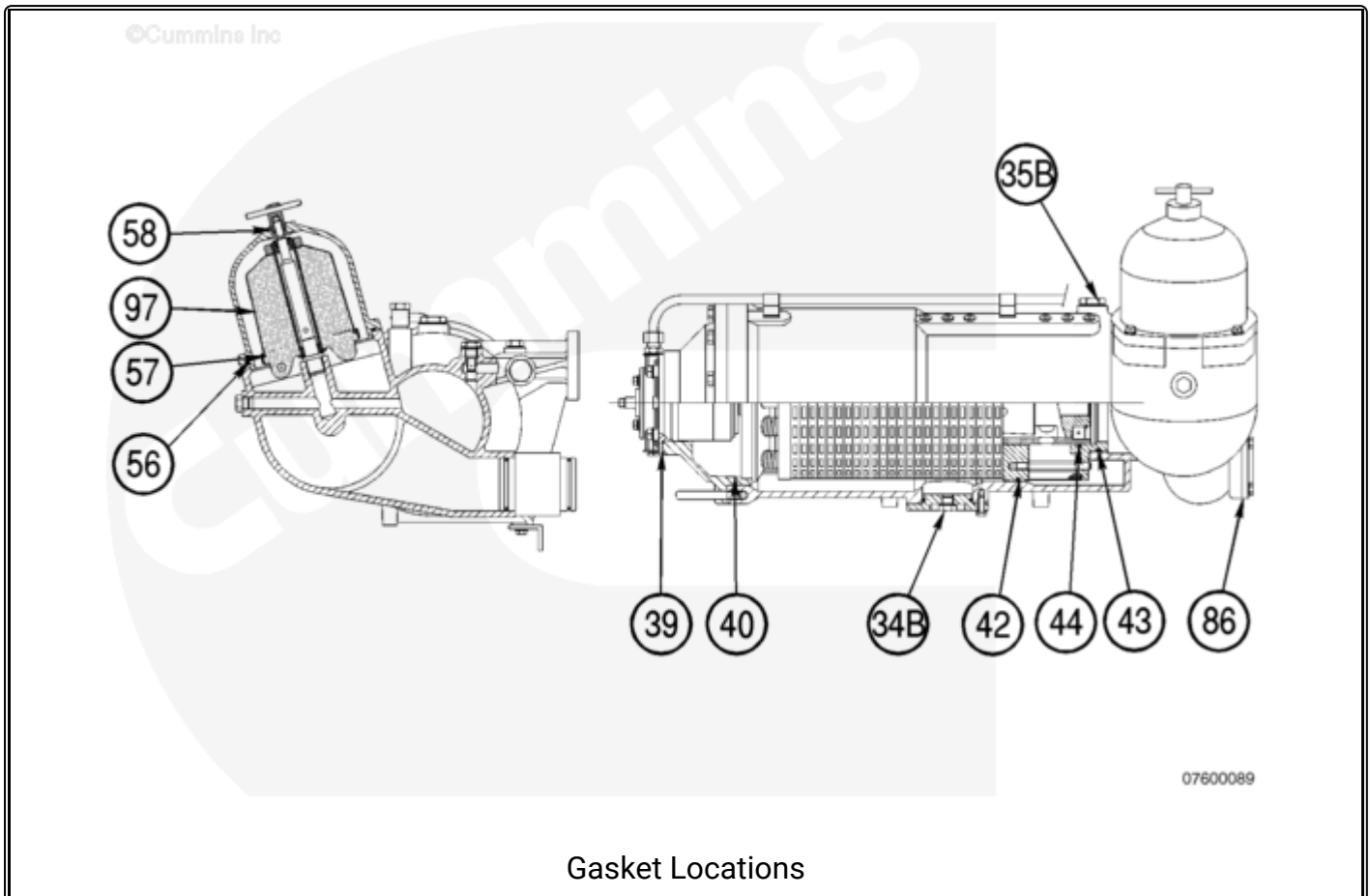
Centrifuge - Exploded View

Centrifuge Parts list

Set of Gaskets for the Filter Housing, Part Number 3631867

Centrifuge			
Item	Description	Part Number	Quantity
55	QSK78 Disposable Centrifuge	3649569	1
55	QST30, QSK38, QSK45, QSK50, QSK60 Disposable Centrifuge	4331734	1
57	O-ring	3393586	1
90	Rotor Body Assembly	-	1

Centrifuge			
Item	Description	Part Number	Quantity
91	Rotor Cover	4016423	1
92	Distributor	3631862	1
93	Top Disc	3631863	1
94	Bowl Disc	3393591	75
95	Distributing Cone	3631864	1
96	Snap Ring	3631865	1
97	Sludge Collecting Paper	-	1



Fleetguard Disposable Centrifuge

A new Cummins Filtration™ one-piece Eliminator™ centrifugal oil filter with SpiraTec™ technology has been released.

For full parts list, reference TSB150164 and TSB160036.

Gasket Parts list

The set of gaskets includes all of the gaskets necessary to do the maintenance of the filter.

This set of gaskets is required every time the filter has a major overhaul, (removal and cleaning of the filtering unit).

Filter Housing Gaskets

Item	Description	Part Number	Quantity
34B	O-ring	3631750	1
35B	O-ring	3631749	1
39	O-ring	3393584	1
40	O-ring	3393565	1
42	O-ring	3393569	1
43	O-ring	3393570	1
44	O-ring	3393571	1
56	O-ring	3393586	1
57	O-ring	3393587	1
58	O-ring	3633001	1
86	O-ring	3631940	1
97	Sludge Collecting Paper	4010887	1

Hydraulic Motor Parts list

Hydraulic Motor

Item	Description	Part Number	Quantity
1000	Hydraulic Motor	6660301	1

Document History

Date	Details
29-sep-2015	Converting service bulletin file type
2015-11-10	Adding type of grease recommendation
2015-12-15	Added inspection criteria to Eliminator™ Filter Maintenance Schedule section.
2017-4-5	Recent Minor Design changes need to identified

Date	Details
2017-4-26	Added note "Eliminator™ Filter spring Springs (25) are required only for Eliminator™ Filters manufactured before September 2016."
2018-3-19	Updating Cummins Part Numbers
2019-2-26	Removed the hydraulic motor internal components and reworded so replace the motor rather than repair.
2021-1-27	Updates to procedures and part numbers.
2022-1-29	Added reference to TSB150164 and TSB160036.
2026-7-24	Updating Cummins Part Numbers

Last Modified: 27-Jul-2026
